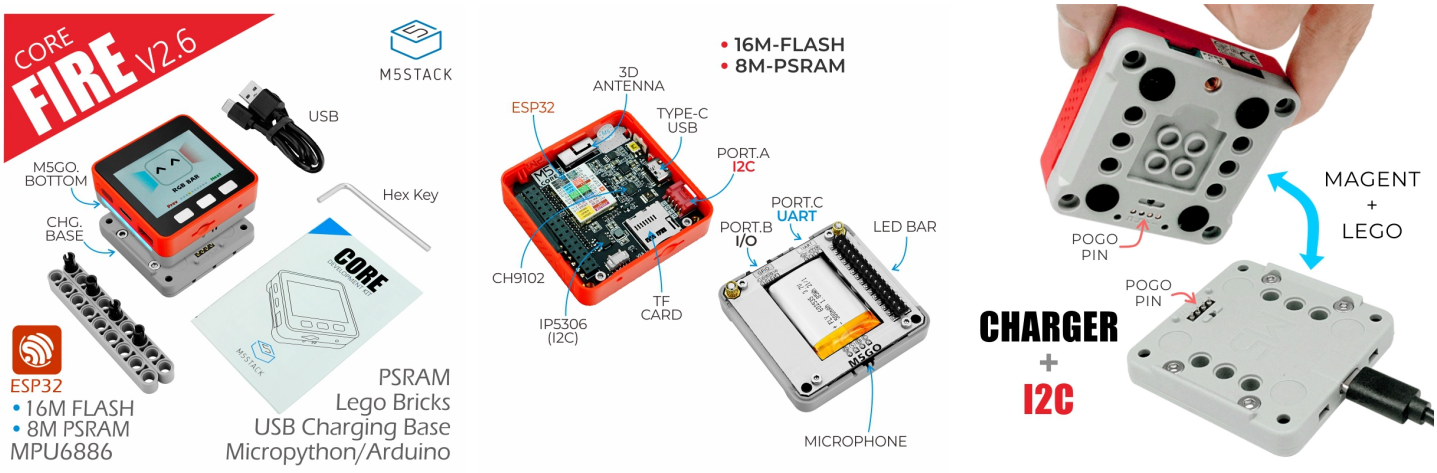
Fire v2.6

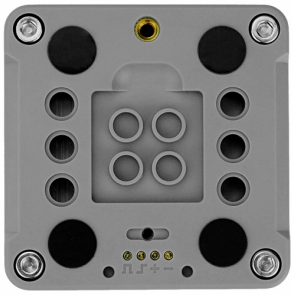
SKU:K007-V26















Description

Fire v2.6 is a cost-effective **Wi-Fi** IoT controller. It uses the Espressif **ESP32** main control chip, equipped with 2 low-power **Xtensa® 32-bit LX6** microprocessors, with a main frequency of up to **240 MHz**. The board comes with **8M PSRAM** + **16M FLASH** memory combination, integrated **2.0-inch full-color HD IPS display panel**, **IMU**, **LED**, **microphone**, **speaker**, **TFCard slot**, and other peripherals. The fully covered

shell ensures **stability** of circuit operation even in complex industrial application scenarios. The internal bus provides a variety of common interface resources (ADC/DAC/I2C/UART/SPI, etc.), with strong expandability. This feature-rich, high-performance IoT controller is very suitable for various product prototype development, industrial control, and smart building application scenarios.

- `Highly Productized`:
  - Exquisite appearance design, prototype development directly corresponds to product implementation
  - Fully covered shell at product level, ensuring more stable circuit operation
- **Low-code Development**:
  - Supports UIFlow graphical programming platform, scripting, no compilation, cloud push
  - Fully compatible with mainstream development platforms such as Arduino, ESP32-IDF, etc.
  - Supports FreeRTOS, leveraging dual-core and multi-task mechanisms to efficiently organize task logic and optimize program execution efficiency
- **High Integration**:
  - 2.0-inch IPS display panel, 6-axis IMU, programmable RGB lights x10, microphone, speaker, custom buttons x3
  - Built-in lithium battery power supply, integrated power management chip, supports Type-C interface and POGO PIN interface power supply
  - Professionally tuned RF circuit, providing stable and reliable wireless communication quality
- **Strong Expandability**:
  - GROVE expansion ports x3 (I2C, GPIO, UART)
  - Easily integrates with M5Stack's software and hardware system, stackable module design, plug-and-play sensor expansion

Tutorial



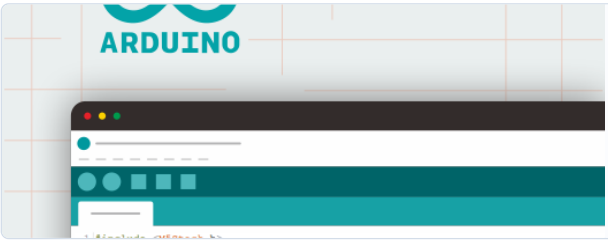
UIFlow

This tutorial will teach you how to control the FIRE device using the UIFlow graphical programming platform



UiFlow2

This tutorial will teach you how to control the FIRE device using the UiFlow2 graphical programming platform



Arduino IDE

This tutorial will teach you how to program and control the FIRE device with the Arduino IDE

Features

- Based on ESP32 development
- 8M PSRAM + 16M Flash memory combination
- Integrated full-color HD IPS display panel and multiple hardware peripherals
- Rich resource interfaces, compatible with M5Stack stackable module system and sensor system, with strong expandability.
- Uses M5GO BOTTOM base compatible with 8mm LEGO bricks, making structure building fun
- Microsoft Azure verified device
- Development Platform
  - UiFlow1
  - UiFlow2

- Arduino IDE
- ESP-IDF
- PlatformIO

Includes

- 1 x M5Stack FIRE
- 1 x M5GO charging base
- 2 x LEGO bricks
- 5 x LEGO connectors
- 1 x M3 hex wrench
- 1 x USB Type-C cable (50cm)
- 1 x User manual

Applications

- IoT controller
- Maker DIY projects
- Smart home control

Specifications

Specification	Parameter
SoC	ESP32-D0WDQ6 @ Xtensa® 32-bit LX6 dual-core processor, clock frequency 240MHz
DMIPS	600
SRAM	520KB
Flash	16MB
PSRAM	8MB Quad
Wi-Fi	2.4 GHz Wi-Fi
Power Input	5V@500mA
Interface	USB Type-C x1, POGO PIN x1, I2C x1, GPIO x1, UART x1
Buttons	Physical buttons x 3
LCD Screen	2.0"@320 x 240 ILI9342C IPS panel, Max brightness 853nit
Speaker	1W-0928
Microphone	Analog BSE3729 Microphone
IMU	6-axis MPU6886
USB Chip	CH9102F
LED	SK6812 RGB LED x 10
Antenna	2.4G 3D antenna
Battery	500mAh@3.7V
Case Material	Plastic (PC)
Product Size	54.0 x 54.0 x 28.6mm
Product Weight	60.9g
Package Size	106.7 x 69.1 x 40.4mm
Gross Weight	143.9g

Learn

BMM150 Magnetic Field Interference

Products with magnets may interfere with the BMM150 magnetic field sensor, causing abnormal readings. When used with an M5 master control device containing a magnet, the magnet needs to be removed, and at the same time, the BMM150 sensor should be kept away from strong magnetic fields.

Power On/Off

Power On/Off Operation

Power On: Click the red power button on the left

Power Off: Quickly double-click the red power button on the left

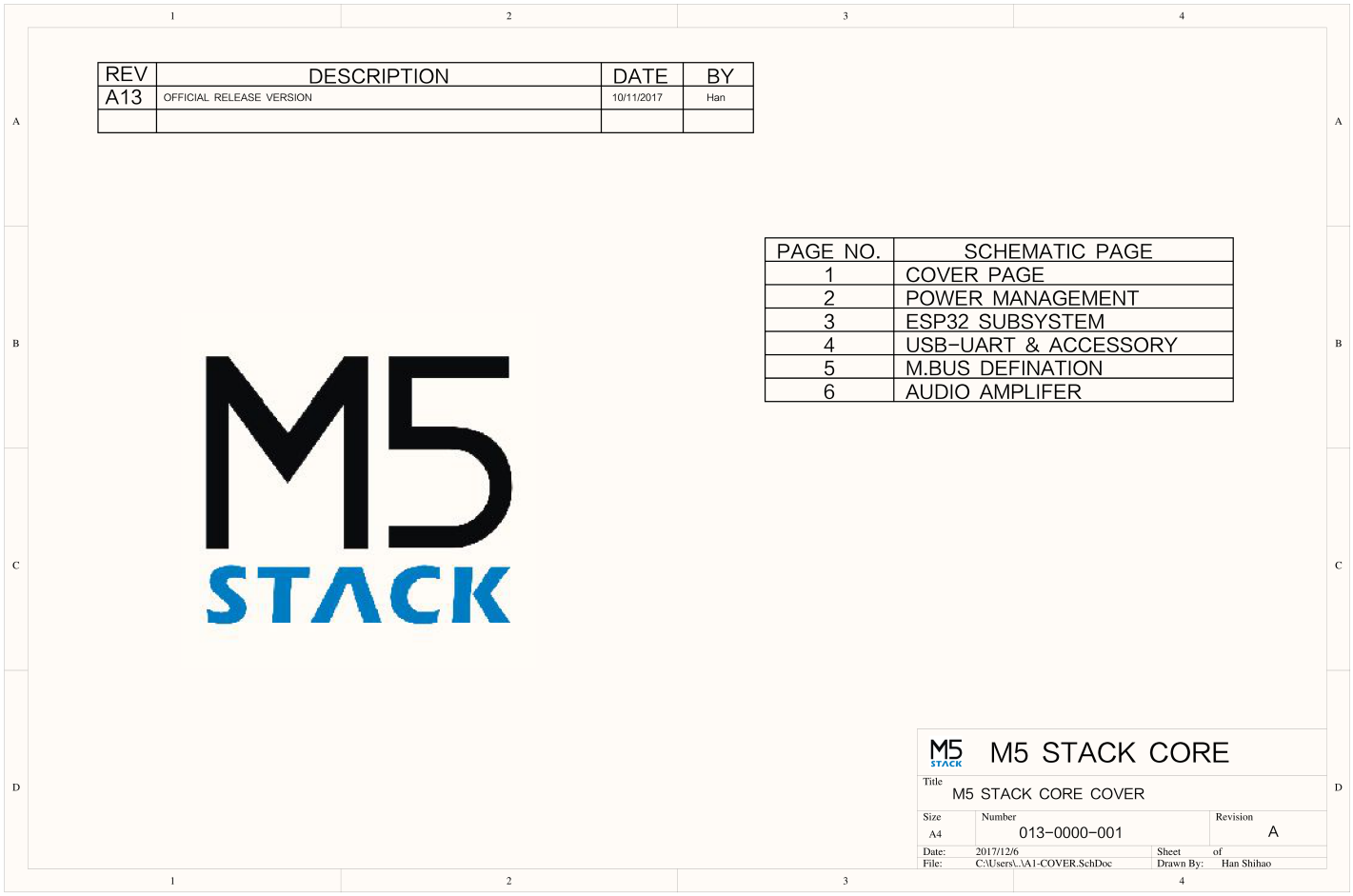
USB Power Supply: By default, when powered by USB, it cannot be turned off

Note:

GPIO 16 / 17 in FIRE are by default connected to PSRAM, so when connecting or stacking other functional modules, be careful to avoid conflicts with these two pins to prevent the device from malfunctioning and causing instability.

Schematics

- M5Core Schematics PDF



REV	DESCRIPTION	DATE	BY
A13	OFFICIAL RELEASE VERSION	10/11/2017	Han

PAGE NO.	SCHEMATIC PAGE
1	COVER PAGE
2	POWER MANAGEMENT
3	ESP32 SUBSYSTEM
4	USB-UART & ACCESSORY
5	M.BUS DEFINATION
6	AUDIO AMPLIFER

M5 STACK

M5 STACK CORE

Title: M5 STACK CORE COVER

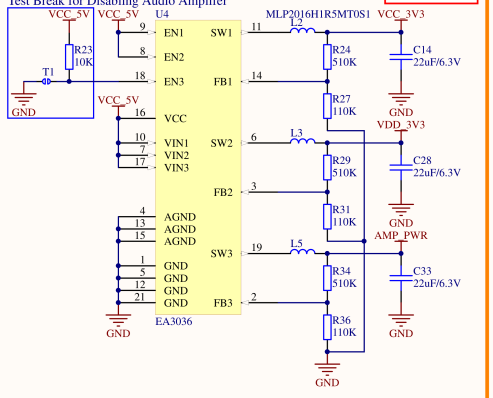
Size	Number	Revision
A4	013-0000-001	A

Date: 2017/12/6 Sheet of

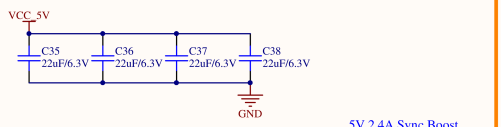
File: C:\Users\..\A1-COVER.SchDoc Drawn By: Han Shihao

Speaker

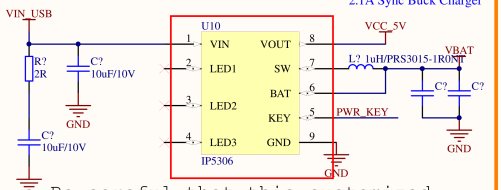
Test Break for Disabling Audio Amplifier



Power



5V 2.4A Sync Boost
2.1A Sync Buck Charger



Be careful that this customized IP5306 was designed to communication with ESP32 through IIC. IIC address is 0x75.

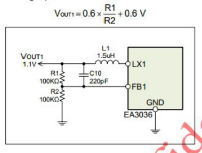
Link for Datasheet: http://198.13.102.98/bj/ingenic_support/X1000_X1000E_X1500/02_HW/00_Halley2/Halley2_coreV3.0_baseV2.0/06Datasheet/PMIC/EA3036_1.0_2014_Sep.pdf

Application Information

Output Voltage Setting

Each of the regulators output voltage can be set via a resistor divider (ex. R1, R2). The output voltage is calculated by following equation:

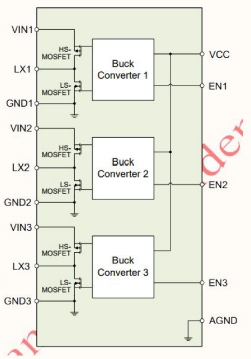
$$V_{OUT1} = 0.6 \times \frac{R1}{R2} + 0.6\ V$$



The following table lists common output voltage and the corresponding R1, R2 resistance value for reference.

Output Voltage	R1 Resistance	R2 Resistance	Tolerance
3.3V	510KΩ	110KΩ	1%
1.8V	200KΩ	100KΩ	1%
1.5V	150KΩ	100KΩ	1%
1.2V	100KΩ	100KΩ	1%

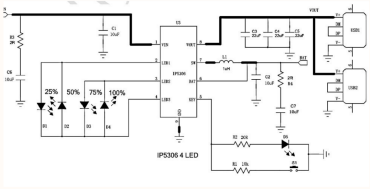
Function Block Diagram



Link for DCDC Buck Inductor: http://www.mouser.com/ds/2/400/nductor_commercial_power_mlp2016_en-838407.pdf

Type		Thickness (mm)max.	L (μH)	tolerance	Measuring frequency (MHz)	DC resistance (Ω)	Rated current (mA)max.	Part No.
Low core loss	Low resistance	1.0	0.47	±20%	2	0.055±25%	1700	MLP2016HF47MT0S1
		1.0	1.0	±20%	2	0.09±25%	1300	MLP2016H1R0MT0S1
		1.0	1.5	±20%	2	0.11±25%	1200	MLP2016H1R5MT0S1
		1.0	2.2	±20%	2	0.11±25%	1200	MLP2016H2R2MT0S1
	Emphasized DC bias characteristics	1.0	3.3	±20%	2	0.12±25%	1200	MLP2016H3R3MT0S1
		1.0	4.7	±20%	2	0.16±25%	1100	MLP2016H4R7MT0S1
		1.0	0.47	±20%	2	0.07±25%	1500	MLP2016V4R7MT0S1
		1.0	1.0	±20%	2	0.12±25%	1200	MLP2016V1R0MT0S1
		1.0	1.5	±20%	2	0.14±25%	1150	MLP2016V1R5MT0S1
		1.0	2.2	±20%	2	0.17±25%	1000	MLP2016V2R2MT0S1

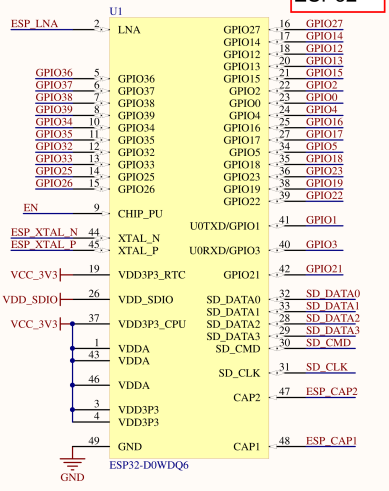
Datasheet for IP5306:
http://www.injoinic.com/doc/IP5306%C2%A0datasheet%C2%A0v1.01.pdf



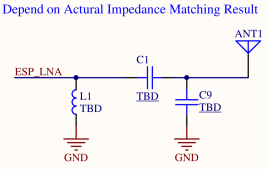
Confidential

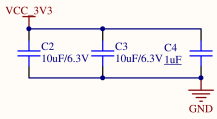
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Revision	A
Date:	2017/12/6
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Drawn By:	Han Shihao

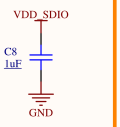
ESP32



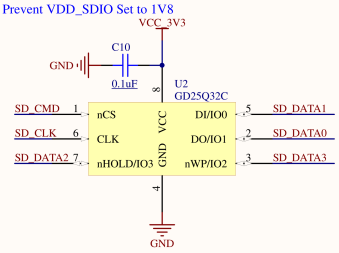
Depend on Actural Impedance Matching Result

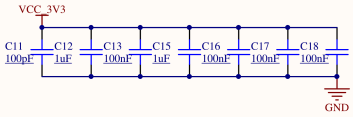




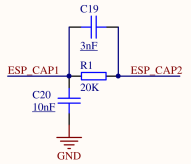


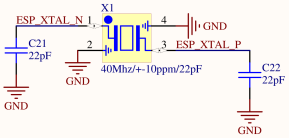
Prevent VDD_SDIO Set to 1V8



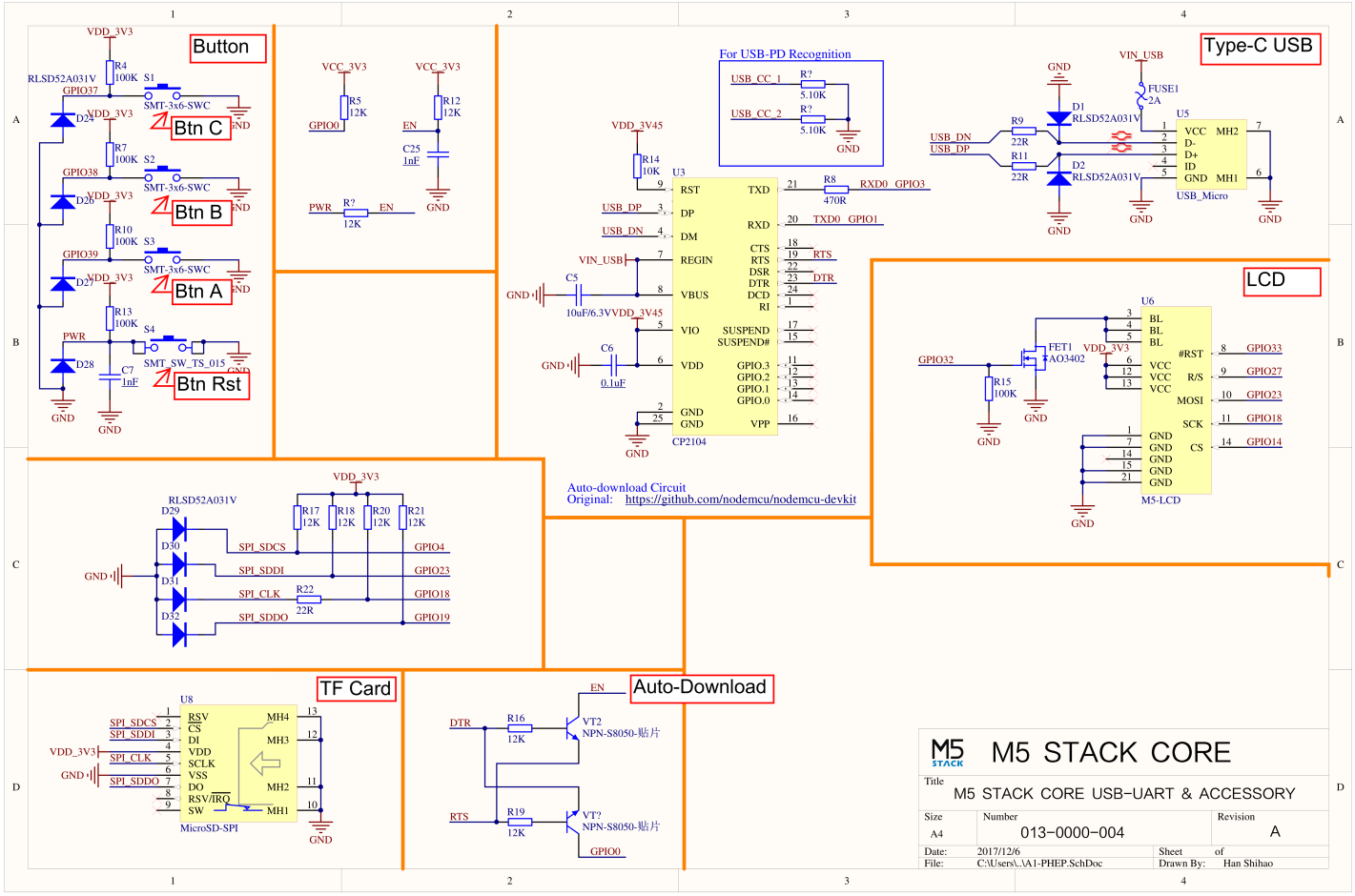


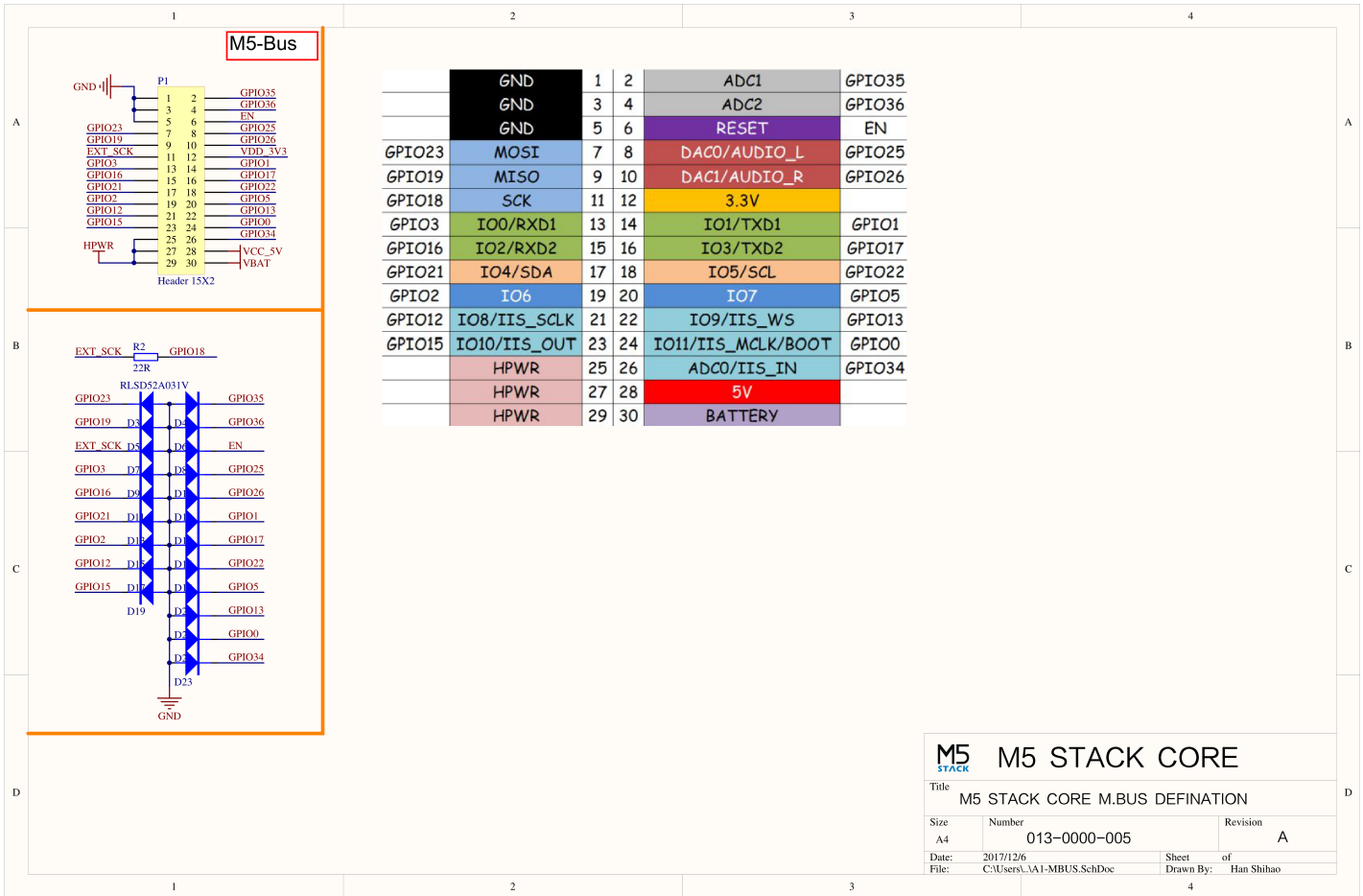
Use 3.3nF for Manufacture





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Size	A4
Number	013-0000-003
Revision	A
Date:	2017/12/6
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File:	C:\Users\..\A1-ESP32.SchDoc
Drawn By:	Han Shihao





	GND	1	2	ADC1	GPIO35
	GND	3	4	ADC2	GPIO36
	GND	5	6	RESET	EN
GPIO23	MOSI	7	8	DAC0/AUDIO_L	GPIO25
GPIO19	MISO	9	10	DAC1/AUDIO_R	GPIO26
GPIO18	SCK	11	12	3.3V	
GPIO3	IO0/RXD1	13	14	IO1/TXD1	GPIO1
GPIO16	IO2/RXD2	15	16	IO3/TXD2	GPIO17
GPIO21	IO4/SDA	17	18	IO5/SCL	GPIO22
GPIO2	IO6	19	20	IO7	GPIO5
GPIO12	IO8/IIS_SCLK	21	22	IO9/IIS_WS	GPIO13
GPIO15	IO10/IIS_OUT	23	24	IO11/IIS_MCLK/BOOT	GPIO0
	HPWR	25	26	ADC0/IIS_IN	GPIO34
	HPWR	27	28	5V	
	HPWR	29	30	BATTERY	

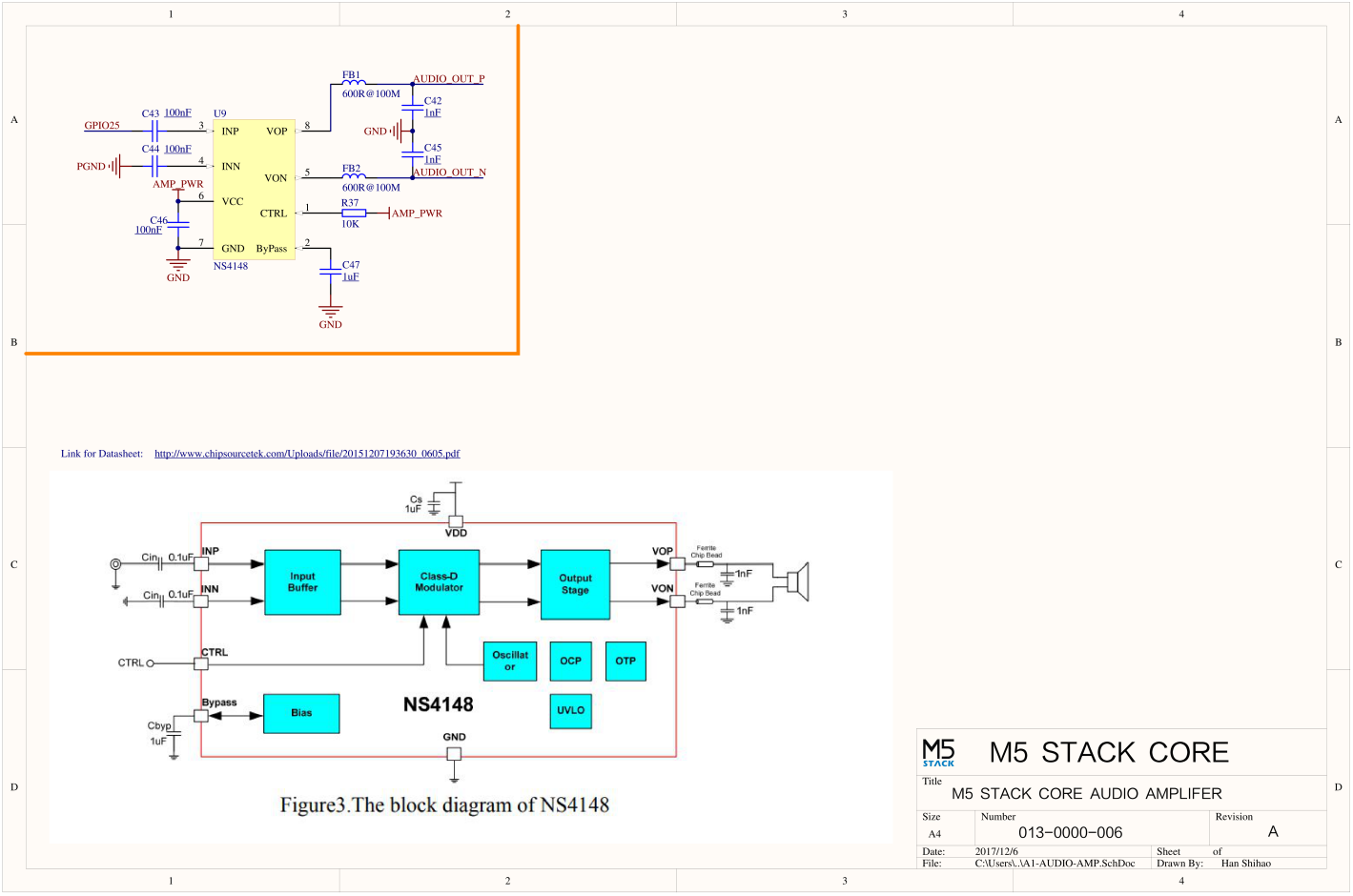


Figure3.The block diagram of NS4148

PinMap

LCD Screen & TF Card

LCD resolution: 320x240 TF card supports up to 16GB

ESP32-D0WDQ6	G23	G19	G18	G14	G27	G33	G32	G4
ILI9342C	MOSI/MISO	/	CLK	CS	DC	RST	BL	
TF Card	MOSI	MISO	CLK	/	/	/	/	CS

Buttons & Speaker

ESP32-D0WDQ6	G39	G38	G37	G25
Button Pins	BUTTON A	BUTTON B	BUTTON C	
Speaker	/	/	/	Speaker Pin

GROVE Interface A & IP5306

The power management chip (IP5306) is a custom I2C version, its I2C address is 0x75. Click here to view the IP5306 register manual.

ESP32-D0WDQ6	G22	G21	5V	GND
GROVE A	SCL	SDA	5V	GND
IP5306 (0x75)	SCL	SDA	5V	GND

IP5306 Charge/Discharge, Voltage Parameters

Charge	Discharge
0.00 ~ 3.40V -> 0%	4.20 ~ 4.07V -> 100%
3.40 ~ 3.61V -> 25%	4.07 ~ 3.81V -> 75%
3.61 ~ 3.88V -> 50%	3.81 ~ 3.55V -> 50%
3.88 ~ 4.12V -> 75%	3.55 ~ 3.33V -> 25%
4.12 ~ / -> 100%	3.33 ~ 0.00V -> 0%

MPU6886 3-axis Accelerometer + 3-axis Gyroscope

MPU6886 I2C address 0x68

ESP32-D0WDQ6	G22	G21	5V	GND
MPU6886 (0x68)	SCL	SDA	5V	GND

M5GO Base PinMap

GROVE Interface

ESP32-D0WDQ6	G36	G26	5V	GND
GROVE B	G36	G26	5V	GND

ESP32 ADC/DAC

ADC1	ADC2	DAC1	DAC2
8 channels	10 channels	2 channels	2 channels
G32-39	G0/2/4/12-15/25-27	G25	G26

HY2.0-4P

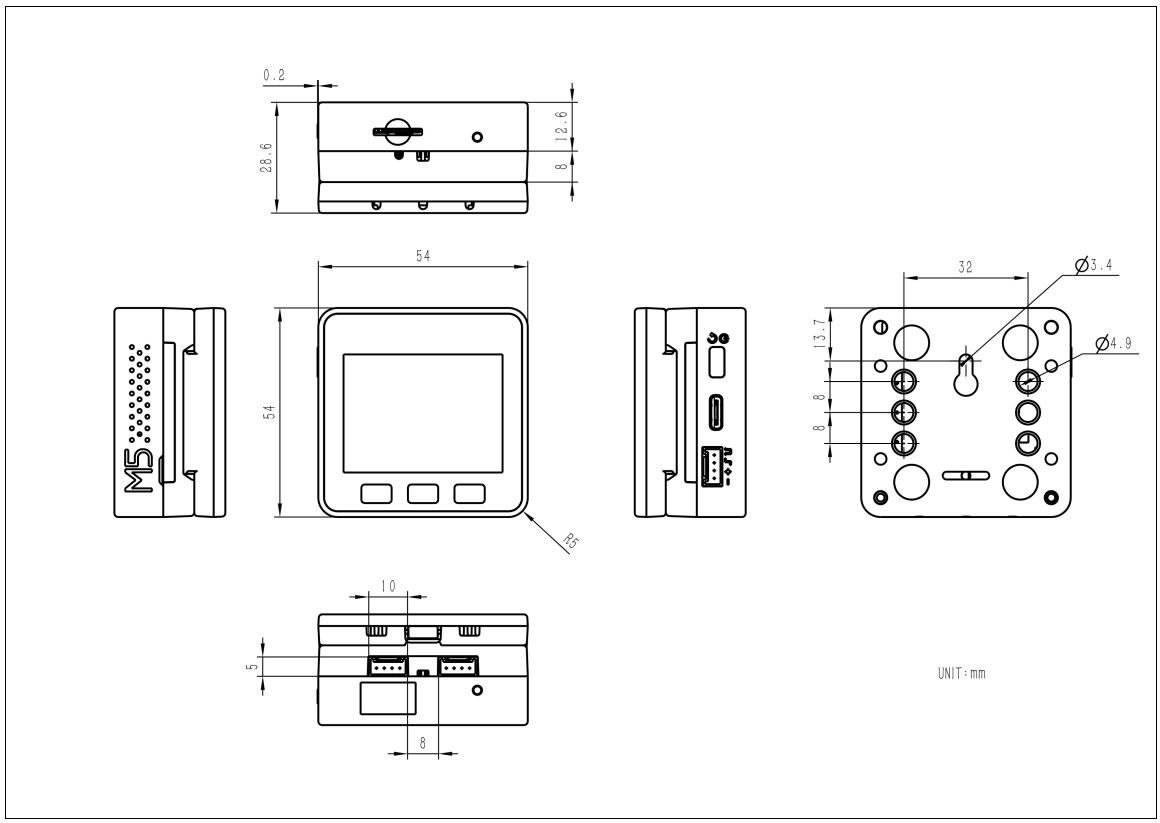
HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	G21	G22
PORT.B	GND	5V	G26	G36
PORT.C	GND	5V	G17	G16

M5-Bus

FUNC	PIN	LEFT	RIGHT	PIN	FUNC
	GND	1	2	G35	ADC
	GND	3	4	G36	ADC
	GND	5	6	RST	EN
MOSI	G23	7	8	G25	DAC/SPK
MISO	G19	9	10	G26	DAC
SCK	G18	11	12	3V3	
RXD0	G3	13	14	G1	TXD0
RXD2	G16	15	16	G17	TXD2
Int SDA	G21	17	18	G22	Int SCL
GPIO	G2	19	20	G5	GPIO
I2S_SK	G12	21	22	G13	I2S_WS
I2S_OUT	G15	23	24	G0	I2S_MK
	HPWR	25	26	G34	I2S_IN
	HPWR	27	28	5V	
	HPWR	29	30	BAT	

When using GPIO15 for RGB LED, it is recommended to initialize the pin with pinMode(15, OUTPUT_OPEN_DRAIN); For more information on pin allocation and pin remapping, please refer to ESP32 datasheet

Model Size



Datasheets

- ESP32
- ILI9342C
- MPU6886
- IP5306

Softwares

Arduino

- Fire v2.6 Arduino Quick Start
- Fire v2.6 Arduino API
- Fire v2.6 Arduino Test Program

UiFlow1

- Fire v2.6 UiFlow1 Quick Start

UiFlow2

- Fire v2.6 UiFlow2 Quick Start

PlatformIO

```
[env:m5stack-fire]
platform = espressif32@6.12.0
board = m5stack-fire
framework = arduino
upload_speed = 921600
monitor_speed = 115200
board_build.partitions = default_16MB.csv
build_type = debug
build_flags =
    -DBOARD_HAS_PSRAM
    -DCORE_DEBUG_LEVEL=5
lib_deps =
    M5Unified=https://github.com/m5stack/M5Unified
```

USB Driver

Click the links below to download the drivers for your operating system. There are currently two versions of driver chips, CP210X (for **CP2104** version)/CP34X (for **CH9102** version) driver packages. After extracting the package, select the installation package corresponding to your operating system's bit version. (If you are unsure which USB chip your device uses, you can install both drivers. **CH9102_VCP_SER_MacOS v1.7** may report an error during installation, but it is actually installed, just ignore it.) If you encounter issues with downloading programs (timeout or Failed to write to target RAM), try reinstalling the device drivers.

Driver Name	Applicable Driver Chip	Download Link
CP210x_VCP_Windows	CP2104	Download
CP210x_VCP_MacOS	CP2104	Download
CP210x_VCP_Linux	CP2104	Download
CH9102_VCP_SER_Windows	CH9102	Download
CH9102_VCP_SER_MacOS v1.7	CH9102	Download

Easyloader

Easyloader	Download Link	Notes
Fire v2.6 Factory Firmware Easyloader	download	/

Video

Product Comparison

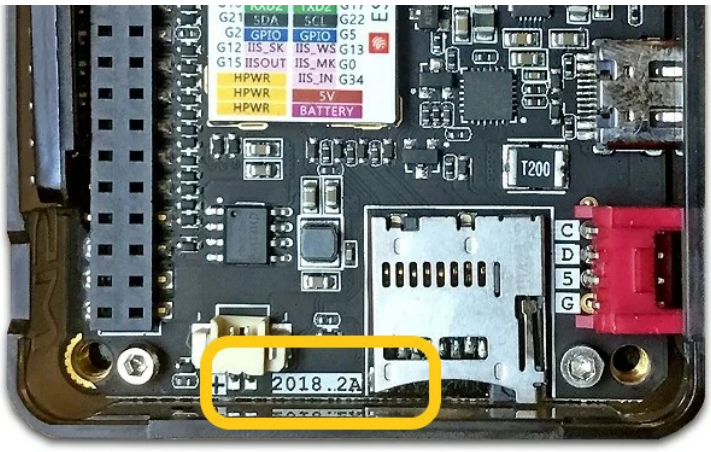
To compare information on the controller series products, you can visit the Product Selection Table, check the target products, and get the comparison results. The selection table covers key information such as core parameters and functional features, and supports comparison of multiple products simultaneously.

Version Change

Release Date	Product Changes	Note
2021.8	Upgraded to v2.6: removed BMM150 magnetometer, changed CP2104 to CH9102, optimized structural details	/
2020.4	PSRAM capacity upgraded from 4MB to 8MB	/
2019.11	Battery capacity adjusted from 600mAh to 500mAh	/
2019.8	Inertial sensor SH200Q changed to MPU6886	/
2019.7	MPU9250 changed to SH200Q+BMM150, TN screen upgraded to IPS screen	Please upgrade your M5Stack library to the latest version (v0.2.8 or above) to solve the screen color inversion problem
2018.6	First release	/

Note

Devices with **2018.2A** PCB version do not support C2C (Type-C to Type-C) connection or PD power supply.


G21 SDA SCL G22
G2 GPIO GPIO G5
G12 IIS_SK IIS_WS G13
G15 IISOUT IIS_MK G0
HPWR IIS_IN G34
HPWR 5V
HPWR BATTERY
T200
C
D
5
G
2018..2A